

Newton's First Law

Grade 8 Physics

Topic: Motion and Force

Guiding question:

Does motion need a force to keep going?

Today we compare observation, experiment, and ideal reasoning.

Everyday Observations

A suitcase slows down after a push.

A skateboard speeds up down a ramp.

A water drop falls toward Earth.

These observations can mislead us if we ignore resistance.

From Aristotle to Galileo

Aristotle: force keeps objects moving.

Galileo: smoother surfaces let objects travel farther.

Ideal reasoning: without resistance, motion can continue uniformly.

Learning Tasks

1. Compare motion on rough and smooth surfaces.
2. Use Galileo's ramp thought experiment.
3. Observe an air-track slider with reduced friction.
4. Explain results with net force.